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Repeatability of Fractal Analysis in OCT Angiography of the Macula and Optic Nerve Head

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Purpose: To evaluate the repeatability of OCT angiography (OCTA) vessel density measurements using fractal analysis in the macula and optic nerve head (ONH) regions.

Design: A prospective longitudinal observational cohort study.

Participants: One hundred sixteen eyes from 71 primary open-angle glaucoma patients with valid macula OCTA scans and 91 eyes from 59 primary open-angle glaucoma patients with valid ONH OCTA scans, with 53 patients contributing scans for both regions.

Methods: Participants underwent 5 OCTA imaging sessions at weekly intervals using the Spectralis spectral-domain OCT (Heidelberg Engineering). OCT angiography images were processed for fractal analysis focusing on small vessels (<25 μm in diameter) within *en face* slabs of the superficial vascular complex (SVC), deep vascular complex (DVC), full retina, and nerve fiber layer vascular plexus (NFLVP). Vessel density repeatability was assessed using the coefficient of variation (CoV) with bootstrapped 95% confidence intervals. Associations between fractal analysis parameters and peripapillary retinal nerve fiber layer (RNFL) thickness were analyzed.

Main Outcome Measures: Repeatability of OCTA vessel density quantified by fractal analysis and correlation with RNFL thickness.

Results: One thousand three hundred twenty-eight macula OCTA scans from 269 visits and 801 ONH scans from 164 visits were included. Fractal analysis demonstrated high repeatability for vessel density measurements, with CoVs ranging from 0.4% to 0.9% in the macula and 0.7% to 1.8% in the ONH region. Full retina slabs exhibited the best repeatability in both regions, while DVC measurements in the ONH showed the highest variability (CoV = 1.8%). Moderate correlations between fractal parameters and RNFL thickness were observed in the ONH (SVC: rho = 0.65; NFLVP: rho = 0.67; $P < 0.001$) and macula (SVC: rho = 0.33, $P < 0.001$).

Conclusions: High repeatability of OCTA vessel density measurements using fractal analysis was observed in the macula and ONH regions, supporting its potential utility in monitoring vascular changes in glaucoma. Future studies should investigate its diagnostic accuracy and ability to detect disease progression.

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Although the pathophysiology of glaucoma is not yet fully understood, there is growing evidence that vascular alterations in the eye may play a significant role in the development and progression of the disease.[1,2] Advances in ophthalmic imaging have opened new pathways to study these vascular factors, with OCT angiography (OCTA) emerging as a powerful tool for assessing ocular blood flow and microvasculature.[3,4] OCT angiography offers a high-resolution, noninvasive imaging approach to visualize the microvascular structures of both the retina and optic nerve head (ONH) across various retinal depths, providing valuable insights into the potential role of vascular dysfunction in glaucoma.[2,5–7]

Various methods have been proposed to assess vascular density in glaucoma patients, with the most common approach being the calculation of vessel density percentage from binarized OCTA images.[8–11] This method typically relies on thresholding techniques to classify vessels based on intensity, providing a straightforward percentage of vascular area relative to the total image area. While useful, percentage-based methods can be limited in sensitivity because they provide only an overall measure of vessel area relative to the total image area. These methods may fail to capture the structural complexity of the network of vessels, such as the branching and connectivity, and may be relatively insensitive to loss of smaller capillaries that may occur in glaucoma.

Fractal analysis, by contrast, offers a more sophisticated way to evaluate the microvascular network by examining its structural complexity and self-similarity.[12,13] In fractal analysis, the microvascular network is represented mathematically as a repeating pattern of branching

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structures, enabling a quantifiable measure of complexity known as the fractal dimension.[12] This metric reflects the density and branching patterns of vessels across the entire network, capturing the inherent irregularity and distribution of vessels, regardless of individual vessel size. Applying fractal analysis to OCTA images may provide significant advantages. Unlike percentage-based methods, fractal analysis can capture subtle alterations in the vascular network, such as reductions in branching complexity or loss of finer vessels, which may be key indicators of early glaucoma progression.[13]

As with any new measurement approach, it is crucial to establish the repeatability of fractal analysis to ensure its reliability and clinical utility. Test–retest repeatability helps to determine whether fractal analysis provides consistent measurements across repeated scans, which is fundamental for its potential application in longitudinal studies and routine clinical practice.

In the present study, we evaluate the repeatability of fractal analysis for quantifying vessel density in OCTA scans in patients with glaucoma, focusing on both the macula and ONH regions. This can provide a foundation to validate these measurements for longitudinal assessment of vessel changes in glaucoma.

Methods

Subjects in this study were selected as part of an ongoing prospective cohort study designed to investigate structural and functional progression in glaucoma. The institutional review board reviewed and approved the present methodology. Participation was voluntary. Each participant read, approved, and signed the written informed consent form. The Declaration of Helsinki and the Health Insurance Portability and Accountability Act were followed throughout the study.

A comprehensive ophthalmic examination was conducted for all participants, including a review of medical history, assessment of visual acuity, slit-lamp biomicroscopy, Goldmann applanation tonometry for intraocular pressure measurement, corneal pachymetry, gonioscopy, dilated fundoscopy examination, stereoscopic optic disc photography, standard automated perimetry using the 24-2 Swedish Interactive Threshold Algorithm Standard on the Humphrey Field Analyzer II model 750 (Carl Zeiss Meditec, Inc), and OCT and OCTA with the Spectralis spectral-domain OCT (SD-OCT) device (Heidelberg Engineering).

All subjects had to be diagnosed with primary open-angle glaucoma before enrollment in the present study. Glaucoma was defined as the presence of $\geq$2 consecutive abnormal standard automated perimetry tests at baseline, characterized by a pattern standard deviation with $P < 0.05$ or glaucoma hemifield test results outside normal limits, and evidence of glaucomatous optic neuropathy. Eyes with concurrent retinal disease, uveitis, non-glaucomatous optic disc neuropathy, spherical refraction beyond $\pm$5.0 diopters, or cylinder correction beyond 3.0 diopters were excluded.

OCTA Testing

To assess the repeatability of OCTA measurements, each participant was scheduled for 5 imaging sessions, spaced approximately 1 week apart. The rationale behind this approach is that any observed variation in measurements can be attributed to test variability rather than disease progression, given that true disease progression would be unlikely within such a short time frame.

During each cluster visit, participants underwent OCTA imaging of both the macula and the ONH regions using the Spectralis SD-OCT. All imaging was performed by the same operator and on the same device. Images for the macula were acquired in the high-resolution mode with 512 B-scans and an angle of 10° centered on the fovea, whereas scans from the ONH region were acquired using the high-speed mode with 512 B-scans and an angle of 20° centered on Bruch membrane opening center. This scan protocol yielded approximately 3 × 3 mm images from the macula and 6 × 6 mm images from the ONH regions.

Image Quality and Processing

The angiographic *en face* slabs were exported for quantification. The following slabs were analyzed for macula: superficial vascular complex (SVC), deep vascular complex (DVC), and full retina. For ONH, slabs were SVC, DVC, full retina, and nerve fiber layer vascular plexus (NFLVP). Sample *en face* images as well as correspondent segmentations for each slab can be appreciated in Figure 1. Images were included if their quality indexes were $\geq$70%. Images with incomplete scanning areas, substantial decentration, or due to motion, blinking, or defocus artifacts, as well as segmentation errors, were excluded from the selection. The same investigator evaluated each image to ensure its quality. Only clusters with a minimum of 4 (maximum of 5) good quality scans were used in the study.

A customized software program was developed to process the *en face* slab images and to perform fractal analysis, as described previously.[7,14,15] Briefly, to create a binary representation of the vessels, the OCTA images underwent processes of inversion, equalization, and removal of nonvessel elements and background noise. The analysis focused on small vessels, defined as those with a diameter of <25 μm, within the SVC and DVC. These microvessels play a critical role in retinal tissue perfusion.[15] The macula images were exported from the device and fed to the software without any mask. For the ONH scans, a circular 1.9 mm-radius mask centered at the Bruch membrane opening was applied automatically before image export to exclude the large vessels of the optic disc from the analysis. The auto-crop function on Adobe Photoshop (Adobe Systems) was used to crop and to resize the images to 1024 × 1024 pixels for the purpose of vessel segmentation. The 8-bit image format was used, displaying 256 shades of gray between the black (0) and white (255). Left eye scans were flipped horizontally for image processing.

For macula scans, the analysis was done within the annular zone, ranging from 0.6 to 2.5 mm, whereas for the ONH region, it was done within the annular zone ranging from 1.9 to 5.0 mm (Fig 2). In fractal analysis, D_{box} (or "box-counting dimension") is a measure of the complexity or "fractality" of an object, in this case, the retinal microvascular network imaged with OCTA. It quantifies how detail in a pattern changes with the scale at which it is measured. For retinal or ONH images, the box-counting method involves overlaying a grid of boxes over the vascular network image and counting the number of boxes that contain any part of the vessels. The grid size is systematically reduced, and the process is repeated. D_{box} is then calculated based on how the number of occupied boxes changes with decreasing box size. A higher D_{box} value suggests a more complex and space-filling network, indicating more branching, connectivity, and density. In the context of OCTA images, a lower D_{box} could indicate a reduction in the density and complexity of the vascular network, which might be

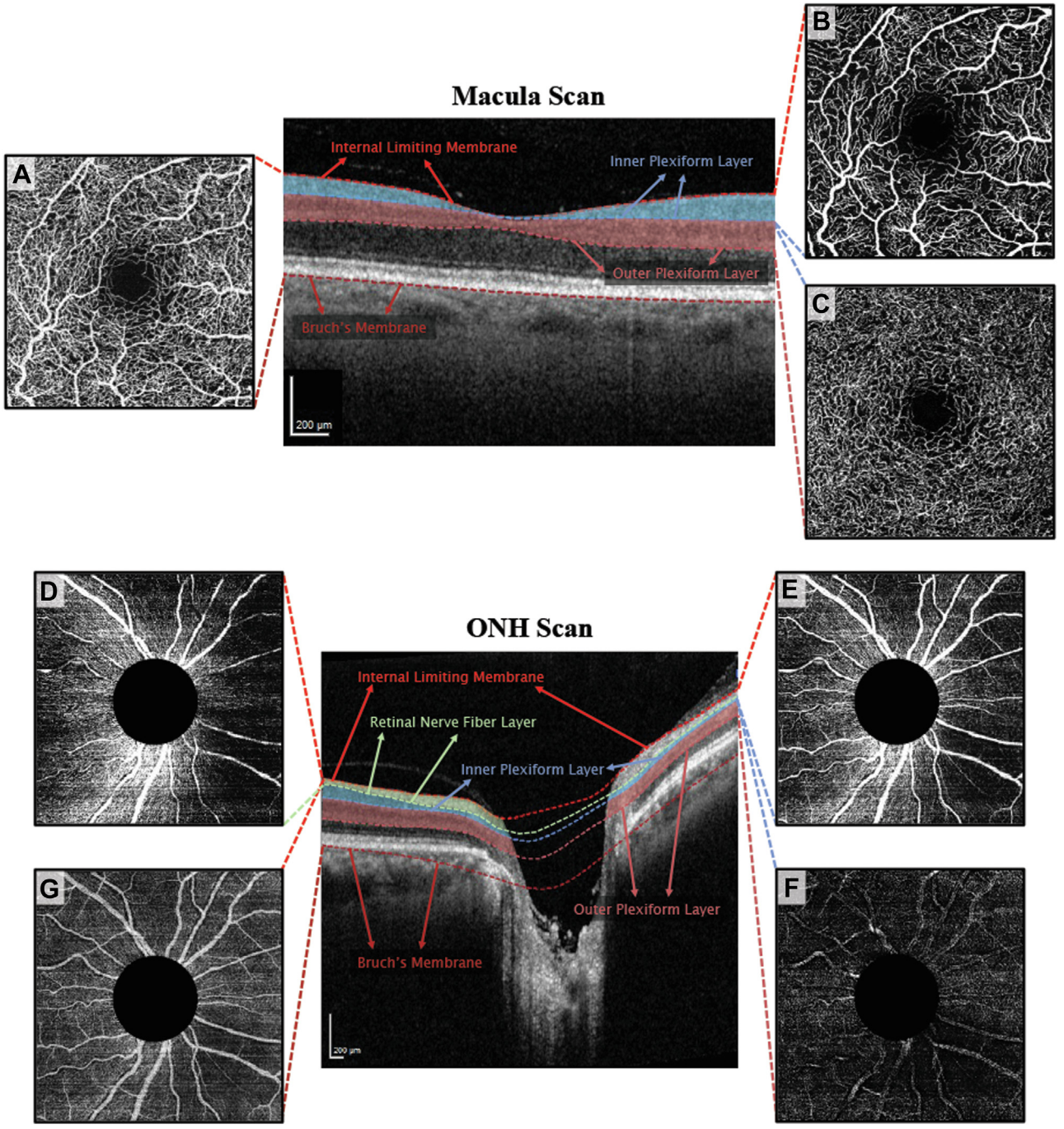


Figure 1. (Top) Schematic diagram of the different layers in a 2-dimensional macula OCT B-scan corresponding to their *en face* OCTA slabs. Examples of macular angiographic *en face* slabs: **(A)** full retina, **(B)** superficial vascular complex, and **(C)** deep vascular complex. (Bottom) Angiographic *en face* slabs for the ONH region and its corresponding regions in a B-scan: **(D)** nerve fiber layer vascular plexus, **(E)** superficial vascular complex, **(F)** deep vascular complex, and **(G)** full retina. Note that the superficial vascular complex corresponds to the analyzed area from the internal limiting membrane to the inner plexiform layer; analogously, the deep vascular complex, from the inner plexiform layer to the outer plexiform layer. Full retina comprehends the portion between the internal limiting membrane and the Bruch membrane. ONH = optic nerve head; OCTA = OCT angiography.

associated with glaucomatous damage. The D_{box} was processed and measured using the fractal analysis toolbox (Benoit Pro version 2.0; TruSoft International, Inc), which applied the box-counting technique with a maximum size of 104 and a rotation of 15 in the fractal analysis settings.[16–19]

Statistical Analysis

The within-cluster repeatability of the D_{box} metric was assessed by the coefficient of variation (CoV) with bootstrapped 95% confidence intervals with 200 repetitions. The standard deviation of the

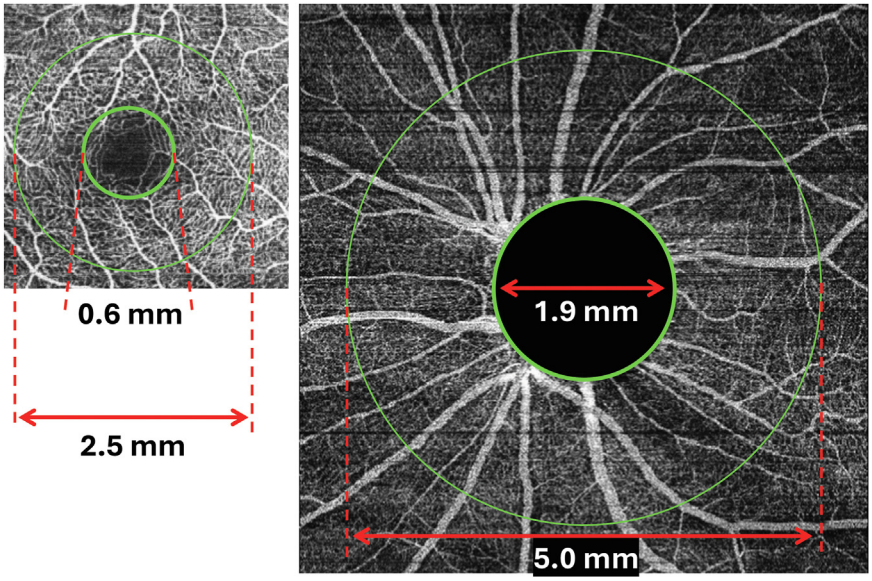


Figure 2. On the left, a representative 3 × 3 mm macula OCTA image and, on the right, a sample 6 × 6 mm ONH OCTA scan of the full retinal vascular network of a research subject. For the ONH, a 1.9 mm diameter circle mask was placed, centered at the Bruch membrane opening. The analyzed area consisted of annular zones (green) from 0.6 to 2.5 mm for the macula and 1.9 to 5.0 mm for the ONH regions. OCTA = OCT angiography; ONH = optic nerve head.

residuals from linear mixed-effects regression models over time, nested at the patient and eye levels for each cluster, was calculated to evaluate the magnitude of the variability of the D_{box} measurements.

Linear mixed models were also used to evaluate the association of peripapillary retinal nerve fiber layer (RNFL) thickness (assessed with SD-OCT) with the CoV to evaluate whether disease severity had an impact in OCTA D_{box} test–retest variability. Statistical analyses were performed using Stata 18 (Statacorp, Inc). The alpha error was set at 0.05.

Results

The study included 116 glaucoma eyes of 71 subjects with valid macula OCTA scans and 91 glaucoma eyes of 59 subjects with valid ONH OCTA scans, with 53 subjects having valid scans for both regions. The analyses included 1328 valid OCTA examinations from 269 clustered visits for the macula region and 801 valid OCTA examinations across 164 clustered visits for the ONH region. A total of 38 macula scans and 112 ONH scans were excluded after review. Table 1 summarizes the reasons for slab image exclusions.

The mean global peripapillary RNFL thickness across all images from SD-OCT was 75.4 ± 19.2 μm (range: 34–123 μm) for the macula region subset and 75.1 ± 19.7 μm (range: 32–123 μm) for the ONH subset. The demographics and clinical characteristics of the subjects included in the study are shown in Table 2.

The average whole retina D_{box} from fractal analysis was 1.731 ± 0.024 in the ONH area and 1.795 ± 0.013 in the macula. In the ONH, there was a statistically significant association between D_{box} values and global RNFL thickness in the SVC (rho = 0.65, $P < 0.001$), NFLVP (rho = 0.67, $P < 0.001$), and full retina (rho = 0.33, $P < 0.001$), but not in the DVC slab (rho = 0.072, $P = 0.387$). Similarly, for the macula scans, lower D_{box} values were associated with thinner RNFL thickness in the SVC (rho = 0.33, $P < 0.001$) and full retina (rho = 0.13, $P = 0.031$), but not in the DVC (rho = 0.03, $P = 0.665$) (Fig 3).

Table 3 shows CoVs and standard deviation of residuals for D_{box} parameters calculated from fractal analysis for each OCTA slab in the ONH and macula regions. The CoV for D_{box} values demonstrated high repeatability across all layers, with low absolute variability for each OCTA *en face* slab. Although small, the CoVs for the ONH region (ranging from 0.67% to 1.80%) were slightly higher than those for the macula, where CoVs ranged from 0.43% to 0.89%. The highest variability was observed in the DVC slabs of the ONH region (1.80%; 95% confidence interval: 1.67%–1.94%). Notably, the full retina OCTA slabs showed the best repeatability in the ONH region.

Table 1. Summary of Excluded Slab Images and Their Respective Causes

Macula Region	ONH Region	Reason	Total Slab Images
24	56	Motion artifacts	80 (53.3%)
8	31	Blinking artifacts	39 (26.0%)
1	21	Segmentation errors	22 (14.7%)
3	4	Defocus	7 (4.7%)
2	0	Decentration	2 (1.3%)

ONH = optic nerve head.

Table 2. Demographic and Clinical Characteristics of the Patients and Eyes Included in the Study

Parameter	Macula	Optic Nerve Head
Eyes/patients (N)	116/71	91/59
Baseline age, yrs range	69.0 ± 7.8 (42.8–82.8)	68.5 ± 8.1 (42.8–79.9)
Sex, %		
Female	42 (59%)	34 (58%)
Male	29 (41%)	25 (42%)
Race, %		
White	50 (70%)	44 (74%)
Black or African American	20 (28%)	15 (26%)
Other	1 (2%)	0 (0%)
Baseline RNFL thickness (μm)	75.4 ± 19.2 (34–123)	75.1 ± 19.7 (32–123)
Overall scan quality index (dB)	36.6 ± 2.8	32.9 ± 2.1
Nerve fiber layer vascular plexus (D_{Box})	-	1.695 ± 0.049 (N = 674)
Superficial vascular complex (D_{Box})	1.740 ± 0.031 (N = 981)	1.704 ± 0.044 (N = 690)
Deep vascular complex (D_{Box})	1.792 ± 0.015 (N = 802)	1.684 ± 0.050 (N = 674)
Full retina (D_{Box})	1.795 ± 0.013 (N = 1252)	1.731 ± 0.024 (N = 721)

dB = decibels; D_{Box} = box-counting dimension; RNFL = retinal nerve fiber layer.
Scan quality index values range from 0 to 40 dB.
Values are shown as mean ± standard deviation, unless otherwise noted.

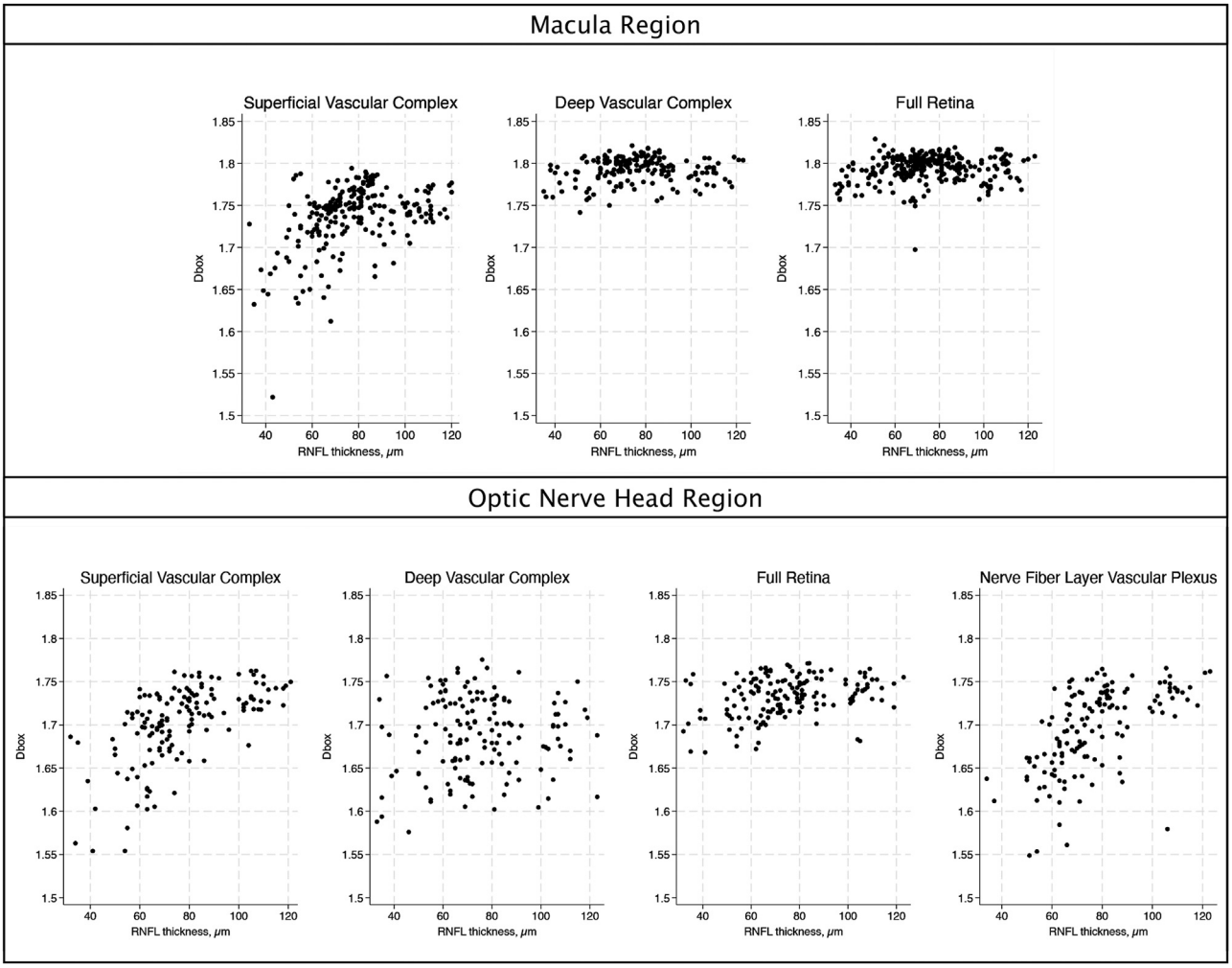


Figure 3. Scatterplots showing the association of the average vessel density values from fractal analysis with structural glaucomatous damage measured by the average RNFL thickness within each cluster for each slab in the macula (top) and optic nerve head (bottom) regions. RNFL = retinal nerve fiber layer.

Table 3. Repeatability Metrics for Vessel Density Based on D_{Box} Values Derived from Fractal Analysis for Both Macular and Optic Nerve Head Regions

	OCTA Slabs	SD of Residuals (Unitless Index)	95% CI	CoV (%)	95% CI
Macula	SVC	0.015	0.014–0.017	0.89	0.80–0.98
	DVC	0.008	0.007–0.009	0.43	0.38–0.49
	Retina	0.008	0.007–0.009	0.44	0.41–0.48
Optic nerve head	SVC	0.016	0.014–0.018	0.94	0.86–1.05
	DVC	0.030	0.028–0.032	1.80	1.67–1.94
	Retina	0.012	0.010–0.013	0.67	0.61–0.75
	NFLVP	0.016	0.015–0.018	0.95	0.85–1.05

CI = confidence interval; CoV = coefficient of variation; D_{box} = box-counting dimension; DVC = deep vascular complex; NFLVP = nerve fiber layer vascular plexus; OCTA = OCT angiography; Retina = full retina; SD = standard deviation; SVC = superficial vascular complex.

When the relationship with disease severity was analyzed (Fig 4), the CoV of vessel density in the ONH was significantly higher in eyes with thinner peripapillary RNFL, specifically in the SVC, full retina, and NFLVP layers, although the increase was modest (0.07%, 0.13%, and 0.14% per 10 μm decrease in RNFL thickness, respectively; all $P < 0.001$). In the macula region, only the full retinal vessel density showed a significant increase in variability with greater glaucomatous damage, with CoV increasing by 0.02% per 10 μm reduction in RNFL thickness ($P = 0.006$).

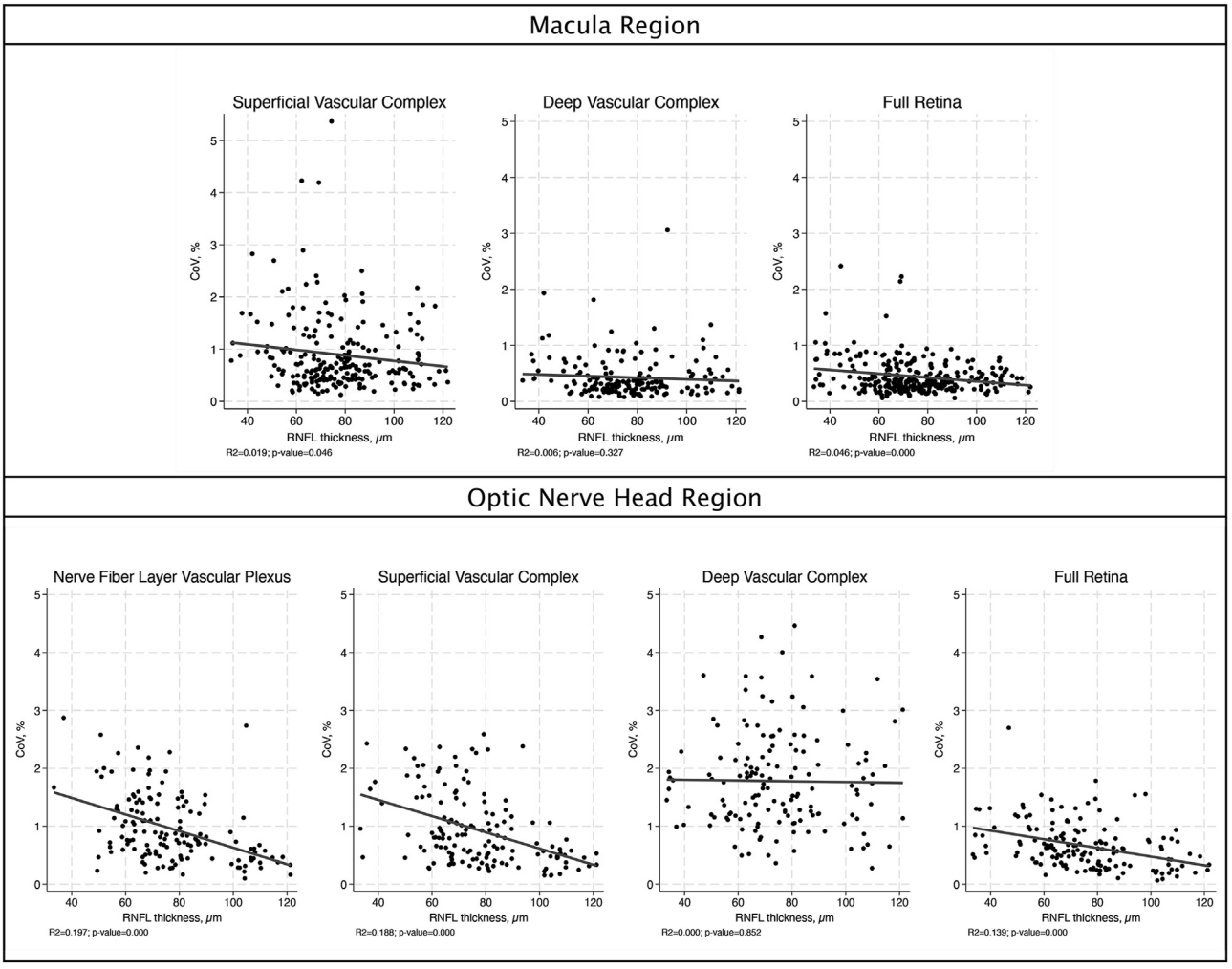


Figure 4. Scatterplots showing the CoV for each slab in the macula and optic nerve head across the spectrum of structural glaucomatous damage, as measured by the RNFL thickness. CoV = coefficient of variability; RNFL = retinal nerve fiber layer.

Discussion

In this study, we evaluated OCTA vessel density repeatability in patients with glaucoma through fractal analysis of >2100 scans of the ONH and macula regions. Our results indicate a high repeatability from OCTA images acquired from Spectralis SD-OCT. For the macula, the mean CoV ranged from 0.4% to 0.9%, and for the ONH, 0.7% to 1.8%, depending on the slab evaluated. These analyses are the largest repeatability report for the macula vessel network using this methodology and are the first to evaluate the ONH region. The high repeatability of OCTA fractal analysis in this study highlights its potential value in diagnosing and detecting glaucoma progression.

Only a few cross-sectional studies with limited samples have explored the macular vessel density using the Spectralis SD-OCT. These studies did not report CoVs, precluding a direct comparison with our study.[20–22] In our analysis, the repeatability of macular vessel density measurements demonstrated CoVs consistently below 1% when employing fractal analysis. Notably, Corvi et al[21] did not evaluate measurement agreement within the same device. Notwithstanding, they observed variations in mean fractal dimension measurements across different models, particularly at the macular deep vascular plexus level.[21]

Fractal analysis is a well-established technique in vascular research, particularly for quantifying the density of large retinal vessels in fundus photographs[16,23,24] and for analyzing retinal microvascular networks in OCTA scans.[7,14,15] This methodology diverges from several earlier studies, which predominantly relied on the automated pixel counting technique for determining vessel density,[9,17,21,25] which is given as the percentage of pixels labeled as vessels in the image. For other devices, such as the Topcon Triton Swept-Source OCT, Fang et al[26] reported a CoV of 0.16% to 0.30% for the full macular vessel density region. Jiang et al[27] observed a CoV of 0.19% to 0.28% for the macula region with the Optovue OCTA device. However, comparing vessel density results across different devices from distinct manufacturers[21] and also among different software versions[18] is challenging. There is also a lack of standardization in the size of the image acquired and significant differences in the segmentation boundaries and position of the scanning area among devices. Additionally, there is no linear relation between vessel density expressed in the percentage of pixels (% area). The present study, therefore, provides cutoffs and expected variability for this technology, which can be further used for the assessment of disease progression in patients with glaucoma.

For vessel density analysis in the peripapillary region, data are even more sparse. Factors such as the presence of large vessels generally cause projection artifacts; different disc anatomy and segmentation challenges due to the presence of the optic disc cup and vessels can hinder objective evaluation of the vessel density in the ONH region.[8,11,28,29] In the present study, we applied a mask centered at the optic disc to try to mitigate that component before applying fractal

Table 4. Repeatability Metrics for OCTA Vessel Density Quantification of the Optic Nerve Head Region in the Scientific Literature

Study	Eyes/Subjects	Device/Region	Scan Protocol	Repeatability Metric	Outcome	Result
Present	91/59 (59 Glaucoma)	**Spectralis SD-OCT** ONH 6 × 6 mm	Clustered visits (5 visits within 6 wks)	CoV, SD Residuals	Fractal dimension	CoV: 0.67%–1.80%, SD: 0.012–0.030
Jia et al, 2014[34]	4/4 (4 Controls)	**Fourier-domain OCT system (custom-made)** ONH 3 × 3 mm	3x+3x+3x, 3 visits	CoV	Disc flow index	CoV: 1.2%–4.2%
Wang et al, 2015[35]	16/16 (16 Controls)	**RTVue-XR Avanti** ONH 3 × 3 mm	2x+2x, 2 visits	CoV, ICC	Vessel density (%)	CoV: 0.7%–3.5%, ICC: 0.80–0.98
Holló, 2016[28]	18/18 (18 Glaucoma)	**Avanti AngioVue.** ONH 4.5 × 4.5 mm	3x+1x scans, 2 visits (3 mo apart)	CoV	Vessel density (%)	CoV_{ONH}: 2.1%–2.9%, $CoV_{RPC\ layer}$: 2.5%–3.8%
Wang et al, 2016[30]	15/15 (15 Controls)	**RTVue-XR OCT** (Optovue Inc, V.2014.2.0.65) ONH 4.5 × 4.5 mm	2x scans, single visit	CoV, ICC	Vessel density (%)	CoV_{ONH}: 1.3%–1.4%, ICC: 0.85–0.94
Venugopal et al, 2017[29]	65/65 (30 Controls, 35 Glaucoma)	**Avanti AngioVue.** ONH 4.5 × 4.5 mm	3x scans, same session	CoV, ICC, Coefficient of repeatability	Vessel density (%)	CoV: 2.5%–2.6%, ICC: 0.86–0.93, Coefficient of repeatability: 4.1%
Manalastas et al, 2017[8]	29/29 (15 Controls, 14 Glaucoma)	**Avanti AngioVue.** ONH 4.5 × 4.5 mm	2x+2x scans, 2 visits (2 wks apart)	CoV, ICC	Vessel density (%)	CoV: 3.8%–4.1%, ICC: 0.55–0.88.
Nishida et al, 2023[11]	88/68 (15 Controls, 38 Glaucoma Suspect and 35 Glaucoma)	**Avanti AngioVue.** ONH 4.5 × 4.5 mm	3 scans on 3 visits within 1–1.5 yrs	CoV, ICC	Vessel density (%)	CoV: 2.5%–2.6%, ICC: 0.823, SD: 1.2%

CoV = coefficient of variation; ICC = intraclass correlation; OCTA = OCT angiography; ONH = optic nerve head; RPC = radial peripapillary capillary; SD = standard deviation; SD-OCT = spectral-domain OCT.

analysis. Our approach had a high repeatability, with CoV below 2% in all layers and as low as 0.7% for the full retina image. It shows improved repeatability compared with the previous few reports of repeatability of ONH vessel density metrics using other devices and vessel segmentation approaches using pixel counting methodology.[8,11,28–30] Table 4 summarizes the OCTA vessel density repeatability assessment of the ONH region present in the scientific literature. These studies have reported a variability ranging from 1.3% to 4.1% using different devices and scan protocols.

In our study, the DVC at the ONH level displayed the largest variation (CoV = 1.80%). This could be explained by the fact that the deeper plexus is more affected by shadow graphic projection artifacts, which may confound the analysis of vessel quantification.[26] We also observed higher variability in both the macula and optic nerve regions with more severe disease. This could be possibly explained by difficulties in segmenting thinner retinal layers, specifically the ganglion cell layer in the macula and the RNFL around the ONH, due to the progressive structural loss happening in glaucoma.[28,31]

We also evaluated whether the vessel density from fractal analysis was correlated with the degree of structural glaucomatous damage, as measured by peripapillary RNFL thickness. In our analysis, the ONH region, particularly the SVC (rho = 0.65) and the NFLVP (rho = 0.67), showed a moderate correlation with RNFL thickness. In contrast, the correlation for the SVC in the macula region was weaker (rho = 0.33, $P < 0.001$). These findings align with previous studies suggesting that OCTA vessel density correlates with glaucoma severity to some extent.[32,33] For instance, Chen et al[32] reported that the density of superficial vessels in the macula had comparable diagnostic precision to peripapillary RNFL and macular ganglion cell complex thickness in distinguishing between glaucomatous and healthy eyes. Our results further revealed that the correlation between RNFL thickness and OCTA fractal analysis parameters is stronger in the ONH region compared to the macula. Future studies should focus on evaluating the diagnostic accuracy of OCTA fractal analysis in detecting glaucoma and monitoring its progression.

A total of 150 slab images were excluded due to quality issues, comprising 38 macula scans and 112 ONH scans. The most common reason for slab exclusion was motion artifacts, accounting for 53.3% of exclusions. These artifacts primarily resulted from involuntary head movements and fixation instability, which can be particularly challenging in patients with advanced glaucoma. Blinking artifacts were the second most frequent cause, contributing to 26.0% of exclusions. These artifacts are often associated with dry eye symptoms, leading to image distortion and data loss. Segmentation errors, affecting 14.7% of excluded images, arose from scan misalignment or software-related issues that compromised automated layer segmentation. Defocus (4.7%) and decentration of the region of interest (1.3%) were additional exclusion factors, typically occurring due to improper scan alignment or patient movement. These findings underscore the inherent challenges of OCTA imaging, particularly in patients with ocular surface disease or advanced glaucoma, where motion instability and segmentation difficulties are more prevalent.

The current study has limitations. First, we did not compare fractal analysis with alternative approaches, such as the pixel-counting method. While both techniques aim to quantify vascular density, their outputs differ fundamentally (% area vs. unitless index), which complicates direct comparisons and may lead to confounding results. Second, we limited our evaluation to mean vessel density measurements within fixed circular patterns around the optic disc or fovea. While this approach offers robust generalizability, smaller, localized sectors may exhibit varying levels of repeatability, and stronger correlations with regional glaucomatous damage. Future research should explore these sectoral variations and assess the diagnostic performance of fractal analysis in detecting localized damage and monitoring disease progression. Additionally, comparisons with alternative measurement techniques are needed to validate the advantages and limitations of fractal analysis in this context.

In conclusion, our results showed high repeatability of the OCTA vessel density quantification using fractal analysis from Spectralis OCTA images in both the macula and ONH vascular network regions. Future studies should determine the value of OCTA vessel density from fractal analysis in diagnosing and detecting glaucoma progression.

Footnotes and Disclosures

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Insurance Portability and Accountability Act were followed throughout the study.

No animal subjects were used in this study.

Author Contributions:

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Data collection: Gameiro, Jammal, Vilasboas-Campos, Wang, Medeiros

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Abbreviations and Acronyms:
CoV = coefficient of variation; **D_{box}** = box-counting dimension; **DVC** = deep vascular complex; **NFLVP** = nerve fiber layer vascular plexus; **OCTA** = OCT angiography; **ONH** = optic nerve head; **RNFL** = retinal nerve fiber layer; **SD-OCT** = spectral-domain OCT; **SVC** = superficial vascular complex.

Keywords:
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References

1. Weinreb RN, Aung T, Medeiros FA. The pathophysiology and treatment of glaucoma: a review. *JAMA*. 2014;311:1901−1911.
2. Sener H, Evereklioglu C, Horozoglu F, Sener ABG. Optic nerve head vessel density using OCTA in patients with primary angle closure disease: a systematic review and network meta-analysis. *Photodiagnosis Photodyn Ther*. 2023;41:103209.
3. Spaide RF, Fujimoto JG, Waheed NK, et al. Optical coherence tomography angiography. *Prog Retin Eye Res*. 2018;64:1−55.
4. Javed A, Khanna A, Palmer E, et al. Optical coherence tomography angiography: a review of the current literature. *J Int Med Res*. 2023;51:3000605231187933.
5. Sampson DM, Dubis AM, Chen FK, et al. Towards standardizing retinal optical coherence tomography angiography: a review. *Light Sci Appl*. 2022;11:63.
6. de Carlo TE, Romano A, Waheed NK, Duker JS. A review of optical coherence tomography angiography (OCTA). *Int J Retina Vitreous*. 2015;1:5.
7. Jiang H, Wei Y, Shi Y, et al. Altered macular microvasculature in mild cognitive impairment and alzheimer disease. *J Neuroophthalmol*. 2018;38:292−298.
8. Manalastas PIC, Zangwill LM, Saunders LJ, et al. Reproducibility of optical coherence tomography angiography macular and optic nerve head vascular density in glaucoma and healthy eyes. *J Glaucoma*. 2017;26:851−859.
9. Salazar-Quiñones L, Peña-Urbina P, Fernández-Vigo JI, et al. Reproducibility of peripapillary, optic nerve head and macular vessel density by OCT-A according to glaucoma severity staging. *Arch Soc Esp Oftalmol (Engl Ed)*. 2023;98:317−324.
10. Kim JA, Lee EJ, Kim TW, et al. Comparison of optic nerve head microvasculature between normal-tension glaucoma and nonarteritic anterior ischemic optic neuropathy. *Invest Ophthalmol Vis Sci*. 2021;62:15.
11. Nishida T, Moghimi S, Hou H, et al. Long-term reproducibility of optical coherence tomography angiography in healthy and stable glaucomatous eyes. *Br J Ophthalmol*. 2023;107:657−662.
12. Jiang H, Debuc DC, Rundek T, et al. Automated segmentation and fractal analysis of high-resolution non-invasive capillary perfusion maps of the human retina. *Microvasc Res*. 2013;89:172−175.
13. Qu D, Lin Y, Jiang H, et al. Retinal nerve fiber layer (RNFL) integrity and its relations to retinal microvasculature and microcirculation in myopic eyes. *Eye Vis (Lond)*. 2018;5:25.
14. Gameiro GR, Gameiro GR, Farah ME, et al. Retinal microvascular density modifications during the water drinking test. *Eur J Ophthalmol*. 2022;32:1602−1609.
15. Lin Y, Jiang H, Liu Y, et al. Age-related alterations in retinal tissue perfusion and volumetric vessel density. *Invest Ophthalmol Vis Sci*. 2019;60:685−693.
16. Tălu S. Multifractal geometry in analysis and processing of digital retinal photographs for early diagnosis of human diabetic macular edema. *Curr Eye Res*. 2013;38:781−792.
17. You Q, Freeman WR, Weinreb RN, et al. Reproducibility of vessel density measurement with optical coherence tomography angiography in eyes with and without retinopathy. *Retina*. 2017;37:1475−1482.
18. Wang H, Hu H, Gregori G, et al. The effect of software versions on the measurement of retinal vascular densities using optical coherence tomography angiography. *Curr Eye Res*. 2021;46:341−349.
19. Jiang H, Gameiro GR, Liu Y, et al. Visual function and disability are associated with increased retinal volumetric vessel density in patients with multiple sclerosis. *Am J Ophthalmol*. 2020;213:34−45.
20. Yang J, Yuan M, Wang E, Chen Y. Comparison of the repeatability of macular vascular density measurements using four optical coherence tomography angiography Systems. *J Ophthalmol*. 2019;2019:4372580.
21. Corvi F, Pellegrini M, Erba S, et al. Reproducibility of vessel density, fractal dimension, and foveal avascular zone using 7 different optical coherence tomography angiography devices. *Am J Ophthalmol*. 2018;186:25−31.
22. Pastore MR, Grotto A, Vezzoni F, et al. Reproducibility and reliability of Spectralis II OCT angiography vascular measurements. *Diagnostics (Basel)*. 2022;12:1908.
23. MacGillivray TJ, Patton N. A reliability study of fractal analysis of the skeletonised vascular network using the "box-counting" technique. *Conf Proc IEEE Eng Med Biol Soc*. 2006;2006:4445−4448.
24. Doubal FN, MacGillivray TJ, Patton N, et al. Fractal analysis of retinal vessels suggests that a distinct vasculopathy causes lacunar stroke. *Neurology*. 2010;74:1102−1107.
25. Levine ES, Arya M, Chaudhari J, et al. Repeatability and reproducibility of vessel density measurements on optical coherence tomography angiography in diabetic retinopathy. *Graefes Arch Clin Exp Ophthalmol*. 2020;258:1687−1695.

26. Fang D, Tang FY, Huang H, et al. Repeatability, interocular correlation and agreement of quantitative swept-source optical coherence tomography angiography macular metrics in healthy subjects. *Br J Ophthalmol*. 2019;103:415–420.
27. Jiang H, Simms AG, Sadaghiani S, et al. Multi-center repeatability of macular capillary perfusion density using optical coherence tomography angiography. *Clin Ophthalmol*. 2022;16:3069–3078.
28. Holló G. Intrasession and between-visit variability of sector peripapillary angioflow vessel density values measured with the angiovue optical coherence tomograph in different retinal layers in ocular hypertension and glaucoma. *PLoS One*. 2016;11:e0161631.
29. Venugopal JP, Rao HL, Weinreb RN, et al. Repeatability of vessel density measurements of optical coherence tomography angiography in normal and glaucoma eyes. *Br J Ophthalmol*. 2018;102:352–357.
30. Wang X, Kong X, Jiang C, et al. Is the peripapillary retinal perfusion related to myopia in healthy eyes? A prospective comparative study. *BMJ Open*. 2016;6:e010791.
31. Liu L, Jia Y, Takusagawa HL, et al. Optical coherence tomography angiography of the peripapillary retina in glaucoma. *JAMA Ophthalmol*. 2015;133:1045–1052.
32. Chen HS, Liu CH, Wu WC, et al. Optical coherence tomography angiography of the superficial microvasculature in the macular and peripapillary areas in glaucomatous and healthy eyes. *Invest Ophthalmol Vis Sci*. 2017;58:3637–3645.
33. Mohammadzadeh V, Moghimi S, Nishida T, et al. Longitudinal structure-function relationship between macular vessel density and thickness and central visual Field in early glaucoma. *Ophthalmol Glaucoma*. 2022;5:648–657.
34. Jia Y, Wei E, Wang X, et al. Optical coherence tomography angiography of optic disc perfusion in glaucoma. *Ophthalmology*. 2014;121:1322–1332.
35. Wang X, Jiang C, Ko T, et al. Correlation between optic disc perfusion and glaucomatous severity in patients with open-angle glaucoma: an optical coherence tomography angiography study. *Graefes Arch Clin Exp Ophthalmol*. 2015;253:1557–1564.